9. The Most General Domains from Integral Transcendental Numbers

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By means of the well-ordering theorem, E. Zermelo constructed a domain of "integral transcendental numbers," that is, a numerical domain $\mathfrak{G}$ having the following abstract properties:

I. The sum, difference, and product of two numbers from $\mathfrak{G}$ is again a number from $\mathfrak{G}$.

II. Every real or complex number is the quotient of two numbers from $\mathfrak{G}$.

III. Every integral rational (respectively algebraic) number is an element of $\mathfrak{G}$.

IV. No non-integral rational (respectively algebraic) number belongs to $\mathfrak{G}$.[1]

The construction rests on the fact that the well-ordering theorem implies the existence of an *algebraic basis* of all numbers: namely, a *system H of numbers* η among which no algebraic relations exist, but by means of which all remaining numbers can be expressed algebraically. Because of their algebraic independence, these basis numbers η can be regarded as indeterminates, and the desired domain therefore becomes an integral domain of rational and algebraic functions of indeterminates η,[2] whose coefficients belong to the field K of all algebraic numbers. The special domain $\mathfrak{G}_\eta$ constructed by Zermelo is then characterized by the following basis properties:

$\mathfrak{G}_\eta$ contains:

1) all basis numbers η,

2) all polynomials in the η with integral[3] coefficients,

3) all integral algebraic functions of the η with integral coefficients.

$\mathfrak{G}_\eta$ excludes:

4) all rationally fractional functions of the η with integral coefficients,

5) all algebraically fractional functions of the η with integral coefficients.[4]

It is now easy to see (compare the examples in §§ 2 and 3) that there are domains $\mathfrak{G}$ essentially different from the Zermelo domain $\mathfrak{G}_\eta$: that is, domains $\mathfrak{G}$ for which, under no well-ordering, do all basis properties 1)–5) hold, and which consequently cannot be generated by Zermelo's construction under any well-ordering. In other words, there are domains $\mathfrak{G}$ that cannot be mapped one-to-one and isomorphically onto $\mathfrak{G}_\eta$. Thus the question arises of the most general domains from integral transcendental numbers; it will be answered completely in what follows.

For this purpose one must first determine which of the basis properties are consequences of the abstract defining properties and which are caused by the special construction. It is shown (§ 1) that for every prescribed domain $\mathfrak{G}$ there is a well-ordering such that basis properties 1) and 2) are satisfied; hence every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis of all numbers. None of the further basis properties need be satisfied (§§ 2 and 3). In particular it follows that a further abstract property of the Zermelo domain does not belong to all domains $\mathfrak{G}$: algebraic integral closedness. It can be formulated as follows:

V. Every number that depends algebraically-integrally and integrally on finitely many numbers from $\mathfrak{G}$ belongs to $\mathfrak{G}$.

The special domains for which V holds in addition to I–IV will be called domains $\mathfrak{H}$ from algebraically-integral transcendental numbers. For the most general domains $\mathfrak{H}$ one obtains the simple results (§ 4):

[1] E. Zermelo, Über ganze transzendente Zahlen, Math. Ann. 75, p. 434 (1914).

[2] For this reason the considerations of the present paper partly run parallel to those of the author's earlier paper: Körper und Systeme rationaler Funktionen, Math. Ann. 76, p. 161 (1915).

[3] Throughout, "integral" is to mean that the coefficients belong to the integral domain $[K]$ of all algebraic integers. For the definition of $\mathfrak{G}_\eta$, it would also suffice in 1)–5) to consider only rational integer coefficients; for more general domains, however, the basis properties would then become less simple.

[4] Zermelo, loc. cit., § 3.

a) The intersection of all domains $\mathfrak{H}$ constructible by means of the same basis H is given by all integral algebraic functions of the basis, that is, by the Zermelo domain $\mathfrak{G}_\eta$, which is itself a domain $\mathfrak{H}$. (This also gives an abstract characterization of the Zermelo domain.)

b) Every domain $\mathfrak{H}$ arises by algebraically-integral adjunction[5] of an arbitrary system $\mathfrak{S}(\eta)$ to $\mathfrak{G}_\eta$, where $\mathfrak{S}(\eta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

The results for the most general domains $\mathfrak{G}$, when an algebraic basis is taken as foundation, are less simple. Here the intersection of all domains $\mathfrak{G}$ is itself not a domain $\mathfrak{G}$, and the construction of the most general domains is consequently harder to survey (§ 5).

In order to obtain results analogous to those for the domains $\mathfrak{H}$, the algebraic basis is replaced by a *rational basis* Θ, that is, by a *system* Θ *of numbers* ϑ which permits all numbers to be expressed rationally, while under at least one well-ordering no basis number can be expressed rationally by finitely many preceding ones. This rational basis Θ must still be subject to the side condition that the integral domain $[\Theta]$ consisting of all integral polynomials in the ϑ contains no algebraically fractional number. (The construction of such a basis with the side condition is shown in § 7.) Then for the most general domains $\mathfrak{G}$ — which one could also call domains from rationally integral transcendental numbers — one again obtains the simple results (§ 6):

a) The intersection of all domains $\mathfrak{G}$ constructible by means of the same rational basis Θ is given by all integral rational functions of the basis, that is, by the domain $[\Theta]$, which is itself a domain $\mathfrak{G}$.

b) Every domain $\mathfrak{G}$ arises by rationally integral adjunction of an arbitrary system $\mathfrak{S}(\vartheta)$ to $[\Theta]$, where $\mathfrak{S}(\vartheta)$ has to satisfy only the single condition that no algebraically fractional number enters the domain through the adjunction.

To have complete analogy with the domains $\mathfrak{H}$, one would have to omit the algebraic numbers from the defining properties III and IV for the domains $\mathfrak{G}$. With this modification of III and IV, the construction of the integral elements by means of a rational basis can be transferred to an arbitrary field (§ 10), whereas Zermelo's construction presupposes the algebraic closedness of the field.

As H runs through every algebraic basis, respectively as Θ runs through every rational basis satisfying the side condition, the results a) and b) give the totality of all domains $\mathfrak{H}$ and $\mathfrak{G}$. If all isomorphic domains are collected into a class, then from this totality one can select a subset which represents the totality of all classes — at least countably infinitely many by § 9 (§ 8).

§ 1. Proof of basis properties 1) and 2) for all domains $\mathfrak{G}$.

Let $\mathfrak{G}$ be any prescribed domain from integral transcendental numbers, so that it has the abstract properties I–IV. Following the procedure applied by E. Zermelo to the field of all complex numbers, we select from $\mathfrak{G}$ an *algebraic basis* H *of all numbers from* $\mathfrak{G}$. Since by II every real or complex number can be represented as the quotient of two numbers from $\mathfrak{G}$, H is at the same time an algebraic basis of *all* numbers. Thus, for every prescribed domain $\mathfrak{G}$, the algebraic basis of all numbers can be chosen so that it belongs to $\mathfrak{G}$; hence basis property 1) is satisfied. Moreover, by III, $\mathfrak{G}$ contains the integral domain $[K]$ of all algebraic integers, and therefore, by I, also all polynomials in the basis elements η with coefficients from $[K]$, that is, all integral polynomials in the η, which form the integral domain $[H]$. Hence basis property 2) is also always satisfied; in other words, every domain $\mathfrak{G}$ can be constructed by means of an algebraic basis H of all numbers in such a way that $\mathfrak{G}$ contains the integral domain $[H]$ as a divisor.

Remark. Nevertheless, starting from a basis H, one can of course construct a domain $\mathfrak{G}$ for which basis properties 1) and 2) do not hold with respect to H itself, but do hold with respect to some other basis H'. For example, let $\mathfrak{G}'$ arise from the Zermelo domain $\mathfrak{G}_\eta$ by replacing, throughout the whole construction, some basis element η by a polynomial $f(\eta)$. Then $\mathfrak{G}'$ contains neither η nor $[H]$; but if in the basis H the basis element η is replaced by $\xi = f(\eta)$ — whereby H again goes over into an algebraic basis H' —, then basis properties 1) and 2) are satisfied for $\mathfrak{G}'$ with respect to H'.

§ 2. Exclusion of basis properties 4) and 5).

We now show that the domains $\mathfrak{G}$ need not satisfy any of the further basis properties. First 4) and 5) are excluded by replacing, in Zermelo's construction, the integral domain $[H]$ by a domain of integral rational

[5] Explanation of the expression in § 4.

"functionals."

According to Weber,[6] a *rational functional* means every rational function with rational number coefficients in the following uniquely determined representation:

$$A(\eta_1 \cdots \eta_t) = a \cdot \frac{E_1(\eta_1 \cdots \eta_t)}{E_2(\eta_1 \cdots \eta_t)},$$

where E_1, E_2 are relatively prime primitive polynomials with rational integer coefficients, and a is a positive rational number (the absolute value of A). If, in particular, a is a *rational integer*, then A is called an *integral rational functional.* As special cases, the integral rational functionals include the rational integers and the polynomials with rational integer coefficients, while rationally fractional numbers and polynomials with rationally fractional coefficients are to be counted among the fractional rational functionals.

Let the domain $\mathfrak{G}$ now consist of the totality $\mathfrak{F}$ of all integral rational functionals of finitely many basis elements η at a time, and of all functions which depend algebraically-integrally on $\mathfrak{F}$ — that is, which satisfy some equation whose leading coefficient is unity and whose remaining coefficients are integral rational functionals.

The proof of properties I–IV for $\mathfrak{G}$ runs exactly parallel to the corresponding proof for Zermelo and will only be indicated briefly. First, II and III are clear, since $\mathfrak{G}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor. By Gauss's theorem on polynomials with rational integer coefficients, sums, differences, and products of integral rational functionals are again integral rational functionals; hence $\mathfrak{F}$ forms an integral domain, and by familiar arguments[7] $\mathfrak{G}$ also becomes an integral domain; thus I is fulfilled. Further, from the extended Gauss theorem[8] follow the two auxiliary propositions: "If z depends algebraically-integrally on $\mathfrak{F}$ by means of *some* equation, then it also does so by means of the *irreducible* equation which z satisfies with respect to the field of all rational functionals,"[9] and "the equation irreducible over the field of all rational numbers and satisfied by an algebraic number is also irreducible over the field of all rational functionals." Thus $\mathfrak{G}$ can contain no fractional algebraic number; hence IV, and therefore all abstract properties I–IV, have been verified.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that $\mathfrak{G}$ excludes all rationally and algebraically fractional functions of the basis elements. For let $z(\eta)$ be any function of the domain to be chosen as a basis element, hence defined by an equation

$$z^k + A_1(\eta)z^{k-1} + \cdots + A_k(\eta) = 0,$$

where

$$A_i(\eta) = a_i \cdot \frac{E_1(\eta)}{E_2(\eta)}$$

are integral rational functionals. Then the function $\frac{a_k}{z}$ belongs to $\mathfrak{G}$, and likewise $\frac{a_k}{\sqrt{z}}$, $\frac{a_k}{\sqrt[3]{z}}$, etc.[10] Hence basis properties 4) and 5) are excluded for the totality of the domains $\mathfrak{G}$.

Remark. An example in § 3 will show that $\mathfrak{G}$ may, under every well-ordering, contain rationally fractional functionals without containing a rationally or algebraically fractional number.

§ 3. Exclusion of basis property 3).

For excluding basis property 3), we use a generalized *module domain* in which the arguments of the polynomials of the domain are no longer the indeterminates, but all integral algebraic functions of the indeterminates, [11] while the indeterminates themselves take over the role of a module basis.

Let $\mathfrak{G}$ consist of *all elements*

$$z = \alpha + g_1(\eta)\eta_1 + g_2(\eta)\eta_2 + \cdots + g_t(\eta)\eta_t, \tag{1}$$

where α runs through all algebraic integers, $\eta_1 \cdots \eta_t$ through all basis elements, and for each fixed combination $\eta_1 \cdots \eta_t$, the functions $g_1(\eta) \cdots g_t(\eta)$ individually run through all integral algebraic functions of the η.[12]

[6] H. Weber, Lehrbuch der Algebra. Kleine Ausgabe §§ 96 and 97.

[7] Compare, for example, Weber, § 97, 8.

[8] Weber, § 20, 5.

[9] Compare Weber, § 97, 4.

[10] Quite analogously, every algebraic function of the η can be transformed into an element of the functional domain $\mathfrak{G}$ by multiplication by a rational integer. Thus this domain has the property that every complex number can be represented as the quotient of an element from $\mathfrak{G}$ and a rational integer, in analogy with the algebraic numbers.

[11] By "integral algebraic functions" are always meant functions with *integral* coefficients; that is, functions satisfying some equation whose leading coefficient is unity and whose remaining coefficients are polynomials in the η with algebraic-integer coefficients — polynomials from $[H]$. In particular, all polynomials from $[H]$ belong to the integral algebraic functions.

[12] The module property belongs to the elements $z - \alpha$.

It is easy to verify that properties I–IV hold for this module domain. First I holds, since the sum, difference, and product of two elements z again takes the form (1). The domain $\mathfrak{G}$ also contains all basis elements η, and besides all elements $\eta \cdot g(\eta)$, where $g(\eta)$ runs through all integral algebraic functions of the η. Hence all integral algebraic functions of the η, and therefore all algebraic functions of the η altogether, can be represented as quotients of two elements of $\mathfrak{G}$, proving II. In the representation (1), all algebraic integers are also included in particular — for $g_1 = 0 \cdots g_t = 0$ —; but z cannot be equal to a fractional algebraic number. For if z represents an algebraic number β, then from

$$\beta = \alpha + g_1(\eta)\eta_1 + \cdots + g_t(\eta)\eta_t$$

identically in the η, necessarily $\beta = \alpha$. This is shown by the specialization

$$\eta_1 = 0 \cdots \eta_t = 0,$$

under which the multipliers $g_1(\eta) \cdots g_t(\eta)$ remain finite as integral algebraic functions, since they go over into integral algebraic functions or numbers.[13] Thus conditions I–IV are satisfied for $\mathfrak{G}$.

On the other hand, under no well-ordering can a basis be selected from $\mathfrak{G}$ in such a way that all integral algebraic functions of the basis elements belong to $\mathfrak{G}$. For let $z(\eta)$ be any function of the domain to be chosen as a basis element — which therefore must actually contain the indeterminates η —. We shall show that, from a certain finite value of ν onward, depending on z, the functions

$$\xi = \sqrt[\nu]{z - \alpha}$$

can no longer belong to $\mathfrak{G}$.

Indeed, suppose that ξ belongs to $\mathfrak{G}$. Then, by (1), it is of the form

$$\xi = \gamma + h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau},$$

where again $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions of the η, and $\eta_{i_1} \cdots \eta_{i_\tau}$ are arbitrary basis elements, which in particular may also coincide with $\eta_1 \cdots \eta_t$. First it must be shown that the algebraic integer γ is zero. Since $h_1(\eta) \cdots h_\tau(\eta)$ are integral algebraic functions, ξ becomes equal to γ for $\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0$ with arbitrary values of the remaining η; in particular, then, ξ is equal to γ for

$$\eta_{i_1} = 0 \cdots \eta_{i_\tau} = 0; \qquad \eta_1 = 0 \cdots \eta_t = 0.$$

But under this specialization the function $(z - \alpha)$ vanishes by (1), and therefore so does ξ; hence $\gamma = 0$, and one has

$$\xi = h_1(\eta)\eta_{i_1} + h_2(\eta)\eta_{i_2} + \cdots + h_\tau(\eta)\eta_{i_\tau}.$$

Raising this to the ν-th power gives

$$\xi^\nu = z - \alpha = F_\nu(\eta),$$

where $F_\nu(\eta)$ denotes a homogeneous, not identically vanishing, form of ν-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are integral algebraic functions of the η. Now $z - \alpha$, which by (1) is an integral algebraic function of the η, satisfies an equation irreducible with respect to $[H]$:

$$(z - \alpha)^\chi + B(\eta)(z - \alpha)^{\chi - 1} + \cdots + C(\eta) = 0,$$

where the last coefficient $C(\eta)$ — the product of $(z - \alpha)$ with its conjugates — is a polynomial; let its degree be λ. On the other hand, from $z - \alpha = F_\nu(\eta)$, by multiplying $F_\nu(\eta)$ by the corresponding conjugates, one obtains

$$C(\eta) = G_{\nu\chi}(\eta),$$

where $G_{\nu\chi}(\eta)$ denotes a homogeneous form of $\nu\chi$-th dimension in $\eta_{i_1} \cdots \eta_{i_\tau}$, whose coefficients are polynomials in the η. Thus $G_{\nu\chi}$ is at least of degree $\nu\chi$ in η, or $\lambda \geq \nu\chi$; that is, the functions $\sqrt[\nu]{z - \alpha}$ for which $\nu > \lambda/\chi$ do not belong to $\mathfrak{G}$. Basis property 3) is therefore excluded for the totality of the domains $\mathfrak{G}$.

Remark. A slight generalization of the domain just constructed leads to a domain $\mathfrak{G}$ which, under every well-ordering, also contains fractional functionals. Let $\mathfrak{G}$ consist of all elements

$$z = \alpha + \gamma_1(\eta)\eta_1 + \gamma_2(\eta)\eta_2 + \cdots + \gamma_t(\eta)\eta_t,$$

where now $\gamma(\eta)$ runs through all algebraically integral, but not necessarily integral, functions of the η. The proof of properties I–IV remains the same as above. But since $\mathfrak{G}$, besides every function z, also contains $a \cdot (z - \alpha)$, where a denotes any rational or algebraically fractional number, fractional functionals also occur under every well-ordering.

[13] This more detailed proof of IV is given because of the expanded example at the end of the paragraph. Here the remark would suffice that, by construction, z is an integral algebraic function.

§ 4. The most general domains from algebraically-integral transcendental numbers.

For a more convenient formulation of what follows, introduce the expression: "an element z depends algebraically-integrally on $\mathfrak{T}$," meaning that z satisfies some equation whose leading coefficient is unity and whose remaining coefficients are integral polynomials in the elements of $\mathfrak{T}$, where $\mathfrak{T}$ denotes any prescribed domain.[14]

A domain $\mathfrak{T}$ is called *algebraically-integrally closed* if it satisfies the condition (compare the introduction):

V. Every element that depends algebraically-integrally on $\mathfrak{T}$ belongs to $\mathfrak{T}$.

We first show that the abstract property V belongs to the Zermelo domain $\mathfrak{G}_\eta$. Let an element z depend algebraically-integrally on $\mathfrak{G}_\eta$ by means of an equation $f(z) = 0$. Then multiplication of $f(z)$ by its conjugates with respect to $[H]$ shows that z also depends algebraically-integrally on $[H]$, and hence belongs to $\mathfrak{G}_\eta$. The algebraically-integral closedness of the functional domain considered in § 2 is shown in the same way.

That property V does not belong to every domain $\mathfrak{G}$ is shown by the module domain considered in § 3, which indeed does not possess basis property 3). More precisely, § 3 showed that the module domain is algebraically-integrally closed with respect to no transcendental element of the domain, whereas, by III and IV, this is of course the case with respect to every algebraic element of the domain.

This leads us first to select, from the totality of domains $\mathfrak{G}$, those which also possess the abstract property V; these are to be called "domains $\mathfrak{H}$ from algebraically-integral transcendental numbers."

Let $\mathfrak{H}$ be such a domain, and let H be an algebraic basis of all numbers from $\mathfrak{H}$; by II, H is at the same time an algebraic basis of all numbers. By III, I, and V, $\mathfrak{H}$ contains, besides H, all integral algebraic functions of the η, that is, the Zermelo domain $\mathfrak{G}_\eta$. Thus, in analogy with § 1: every domain $\mathfrak{H}$ from algebraically-integral transcendental numbers can be constructed by means of an algebraic basis of all numbers in such a way that $\mathfrak{H}$ contains the Zermelo domain $\mathfrak{G}_\eta$ as a divisor.

Now let H be any algebraic basis of all numbers, and let $\mathfrak{M}(H)$ denote the totality of all domains $\mathfrak{H}$ containing H. Every domain $\mathfrak{H}$ in $\mathfrak{M}(H)$ contains, as just proved, the Zermelo domain $\mathfrak{G}_\eta$ as a divisor; and since $\mathfrak{G}_\eta$ itself is a domain $\mathfrak{H}$, $\mathfrak{G}_\eta$ is the greatest common divisor, or intersection, of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$. Thus $\mathfrak{G}_\eta$ has been defined abstractly. It also follows directly that $\mathfrak{M}(H)$ is closed under intersection: that is, the intersection of all domains $\mathfrak{H}$ of any subsystem of $\mathfrak{M}(H)$ again belongs to $\mathfrak{M}(H)$. For, as an intersection, I and V are satisfied; II and III follow because $\mathfrak{G}_\eta$ is a divisor; IV follows because the intersection is a divisor of $\mathfrak{H}$.

As the example of the functional domain in § 2 shows, the domains $\mathfrak{H}$ in general contain further functions besides $\mathfrak{G}_\eta$. Let $\psi(\eta)$ be such a rational or algebraic function of the η. Then, by I, $\mathfrak{H}$ also contains all integral rational combinations of ψ and finitely many functions from $\mathfrak{G}_\eta$ at a time. The integral domain so arising — consisting of all polynomials in ψ with coefficients from $\mathfrak{G}_\eta$ — will be denoted by $[\mathfrak{G}_\eta, \psi]$, and the process leading to it will be called "rationally integral adjunction of ψ to $\mathfrak{G}_\eta$." By V, $\mathfrak{H}$ further contains all functions which depend algebraically-integrally on $[\mathfrak{G}_\eta, \psi]$; the domain so arising will be denoted by $\{\mathfrak{G}_\eta, \psi\}$, and the process leading to it by "algebraically integral adjunction of ψ to $\mathfrak{G}_\eta$." The domain $\{\mathfrak{G}_\eta, \psi\}$ consists of all integral algebraic functions of ψ with coefficients from $\mathfrak{G}_\eta$, and is therefore an integral domain by familiar arguments.[15]

We show that $\{\mathfrak{G}_\eta, \psi\}$ is also algebraically-integrally closed, that is, satisfies condition V. Indeed, let z depend algebraically-integrally on $\{\mathfrak{G}_\eta, \psi\}$ by means of an equation

$$f(z; a \cdots c) = z^\nu + az^{\nu-1} + \cdots + c = 0,$$

where $a \cdots c$, as elements of $\{\mathfrak{G}_\eta, \psi\}$, satisfy equations with coefficients from $[\mathfrak{G}_\eta, \psi]$:

$$a^\lambda + A_1 a^{\lambda-1} + \cdots + A_\lambda = 0,$$

$$\vdots$$

$$c^\mu + C_1 c^{\mu-1} + \cdots + C_\mu = 0.$$

Let their roots be denoted by $a, a' \cdots a^{(\lambda-1)} \cdots c, c' \cdots c^{(\mu-1)}$. If one now forms the product over all combinations of the roots,

$$g(z) = f(z; a \cdots c) f(z; a' \cdots c') \cdots f(z; a^{(\lambda-1)} \cdots c^{(\mu-1)}),$$

[14] For an integral domain $\mathfrak{T}$, the leading coefficient is unity and the remaining coefficients are elements of $\mathfrak{T}$.

[15] Compare, for example, Weber, § 97, 6 and 7.

then $g(z)$ has unity as leading coefficient and elements of $[\mathfrak{G}_\eta, \psi]$ as its remaining coefficients. Consequently z belongs to $\{\mathfrak{G}_\eta, \psi\}$.[16]

Thus the domain $\{\mathfrak{G}_\eta, \psi\}$ has properties I and V; and, since $\mathfrak{G}_\eta$ is a divisor of the domain, it also has II and III. Whether IV is satisfied depends on the choice of ψ; for example, IV is not satisfied for $\psi = \frac{1}{n \cdot \eta}$, since then $\frac{1}{n}$ belongs to the domain. It is satisfied, however, by § 2 for $\psi = \frac{1}{\eta}$, or also for $\psi = \frac{\eta}{n}$. For in the latter case, if a function from $\{\mathfrak{G}_\eta, \psi\}$ represents an algebraic number, its value remains unchanged for $\eta = 0$; thereby it passes into a function from $\mathfrak{G}_\eta$, and consequently into an algebraic integer.

Thus $\{\mathfrak{G}_\eta, \psi\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on ψ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction.

Quite analogously one defines the rationally integral, respectively algebraically integral, adjunction of an arbitrary system $\mathfrak{S}$ of rational or algebraic functions of the η to $\mathfrak{G}_\eta$. The rationally integral adjunction leads to the integral domain $[\mathfrak{G}_\eta, \mathfrak{S}]$, consisting of all integral polynomials in finitely many elements from $\mathfrak{G}_\eta$ and $\mathfrak{S}$ at a time. The algebraically integral adjunction gives the domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ consisting of all elements that depend algebraically-integrally on $[\mathfrak{G}_\eta, \mathfrak{S}]$. Exactly as above, one shows that the integral domain $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ is algebraically-integrally closed and also satisfies II and III. Thus, as above:

$\{\mathfrak{G}_\eta, \mathfrak{S}\}$ becomes a domain $\mathfrak{H}$ as soon as one imposes on $\mathfrak{S}$ the condition that no algebraically fractional number enters the domain through the algebraically integral adjunction — that is, as soon as $\mathfrak{S}$ becomes an admissible system.[17]

It is now important that the converse also holds, so that the domains $\{\mathfrak{G}_\eta, \mathfrak{S}\}$ actually exhaust all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ as $\mathfrak{S}$ runs through all admissible systems. Indeed, let $\mathfrak{H}$ be any domain from $\mathfrak{M}(H)$ and denote by $\mathfrak{L} = \mathfrak{H} - \mathfrak{G}_\eta$ the system consisting of all functions from $\mathfrak{H}$ not belonging to $\mathfrak{G}_\eta$. By V, $\mathfrak{H}$ contains the domain $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ as a divisor; but $\{\mathfrak{G}_\eta, \mathfrak{L}\}$ is also divisible by $\mathfrak{H}$, because it contains $\mathfrak{G}_\eta$ and $\mathfrak{L}$, and since these have no common element, it contains $\mathfrak{H}$ as well. Hence $\mathfrak{H} = \{\mathfrak{G}_\eta, \mathfrak{L}\}$; and since IV is satisfied for $\mathfrak{H}$, $\mathfrak{L}$ is of course an admissible system.[18]

As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs, as proved at the beginning, through the totality of all domains $\mathfrak{H}$. Consequently the question of the most general domains $\mathfrak{H}$ from algebraically-integral transcendental numbers is completely answered by the results of this paragraph:

a) The intersection of all domains $\mathfrak{H}$ from $\mathfrak{M}(H)$ is given by the Zermelo domain $\mathfrak{G}_\eta$, which itself is a domain $\mathfrak{H}$; $\mathfrak{M}(H)$ is closed with respect to forming intersections.

b) Every domain $\mathfrak{H}$ from $\mathfrak{M}(H)$ arises by algebraically integral adjunction of an admissible system $\mathfrak{S}$ to $\mathfrak{G}_\eta$. As H runs through every possible algebraic basis, $\mathfrak{M}(H)$ runs through the totality of all domains $\mathfrak{H}$.[19]

[16] Thus, by defining algebraic integrality by means of *some* equation, the concept of irreducibility with respect to a ground field becomes unnecessary for the proof of I and V. Compare also § 2, where, only for the proof of IV, the auxiliary proposition is needed which passes from the definition of algebraic integrality by an arbitrary equation to the irreducible equation. Since this auxiliary proposition no longer holds in general in the present case, IV has to be imposed on the function ψ as a special condition.

[17] More generally, one shows analogously: if $\mathfrak{T}$ is any domain and $\{\mathfrak{T}\}$ denotes the totality of elements depending algebraically-integrally on $\mathfrak{T}$, then $\{\mathfrak{T}\}$ is algebraically-integrally closed.

[18] The adjunction of a subsystem of $\mathfrak{L}$ can already suffice; for example, the functional domain of § 2 arises by algebraically integral adjunction of all integral rational functionals — with the exception of the polynomials — to $\mathfrak{G}_\eta$.

[19] The admissible systems $\mathfrak{S}$ can be found as follows. Put all algebraically fractional functions of the η under some well-ordering Ω, and take into $\mathfrak{S}$ all functions except those which, when algebraically-integrally adjoined to $\mathfrak{G}_\eta$ and the previously admitted functions, generate an algebraically fractional number. This procedure can be broken off at any arbitrary place. As Ω runs through every well-ordering, and as the process is broken off at every arbitrary place for each fixed Ω, all admissible systems $\mathfrak{S}$ are thereby exhausted.